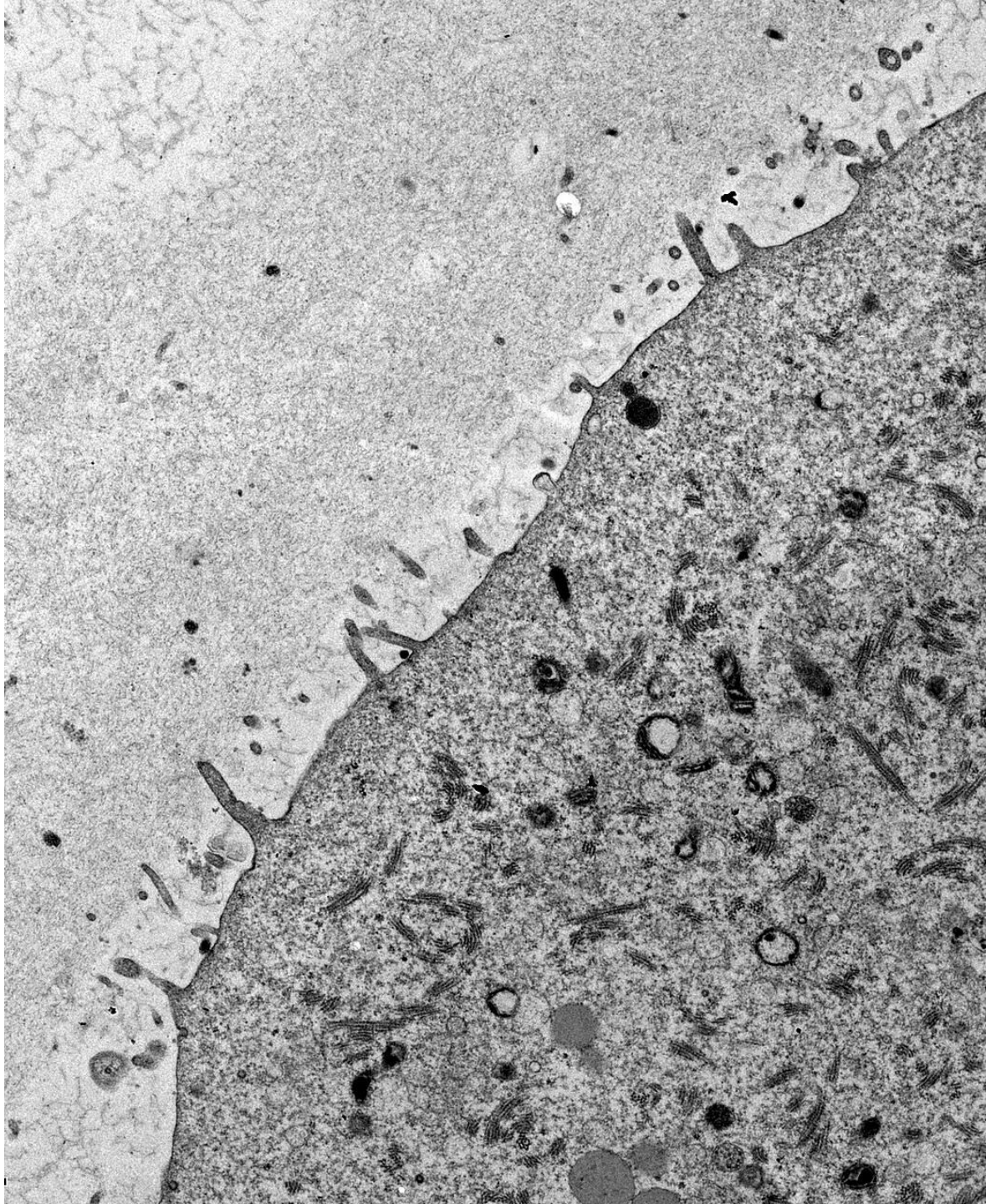
DP Biology – D3.1

Reproduction

- What does sexual or asexual reproduction show about change and continuity?
- How must organisms change in order to reproduce?

Statements D3.1.1–7, SL and HL. Textbook pp. 693–702. Plants are in Plant Systems.
Image: the surface of an egg and its outer coat, electron micrograph. Prue Talbot, CC0.

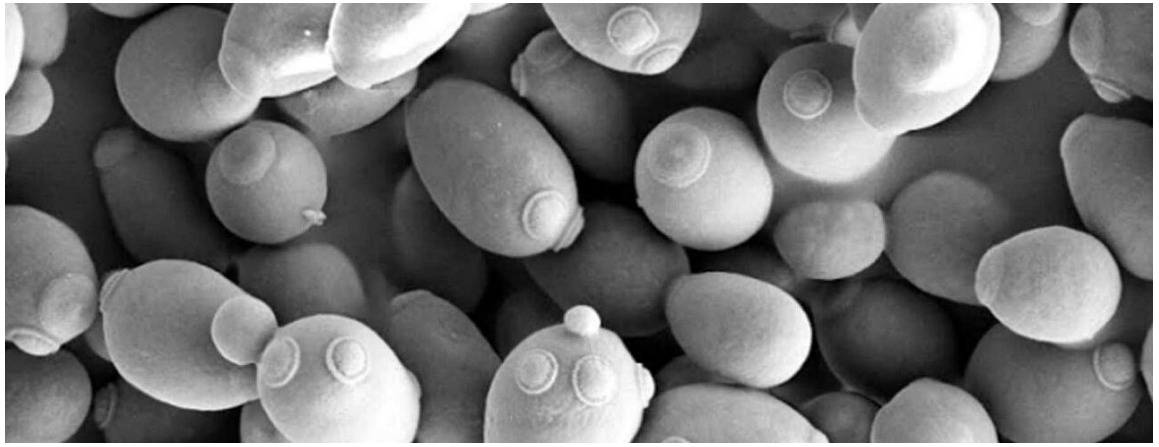


Objectives for each class

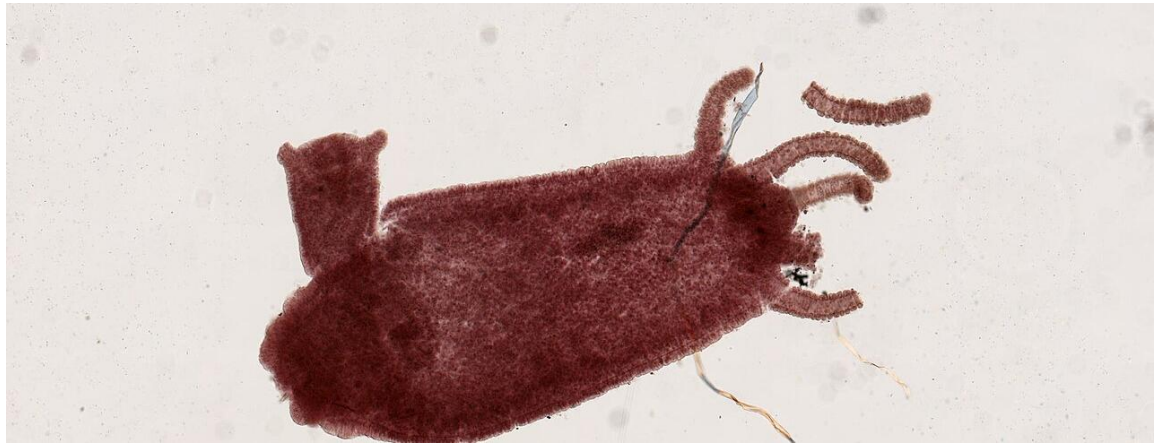
Class	Statements	By the end of the class, you can...
1	D3.1.1–3	Compare sexual and asexual reproduction, and explain the roles of meiosis, fusion and the two kinds of gamete
2	D3.1.4–5	Annotate the reproductive systems, and explain the hormonal control of the menstrual cycle
3	D3.1.6–7	Outline fertilization, and explain and evaluate the use of hormones in IVF

Asexual reproduction: one parent

D3.1.1



Yeast cells budding



Hydra budding

Images, left to right: M.D. Murtey and P. Ramasamy, CC BY 3.0; A. Gomez-Colunga, CC0.

No gametes: cells divide by **mitosis**, so offspring are **genetically identical** to the parent. No partner is needed and numbers rise fast.

Advantage

Identical offspring stay well adapted, as long as the environment stays the same.

Sexual reproduction: two parents

D3.1.1



Trout spawning. Parker Montgomery, CC BY-SA 3.0.

Meiosis makes haploid gametes; **fertilization** joins two into a diploid zygote. Each offspring has a **new combination** of genes.

It costs more time and energy, and usually needs a partner.

Advantage

Variation means some offspring may be better suited if the environment changes.

Sexual and asexual compared

D3.1.1

	Asexual	Sexual
Parents	One	Usually two
Gametes	None	Haploid gametes
Offspring	Genetically identical	Genetically varied
Suited to	A stable environment	A changing environment
Examples	Bacteria, yeast, hydra	Most animals, humans

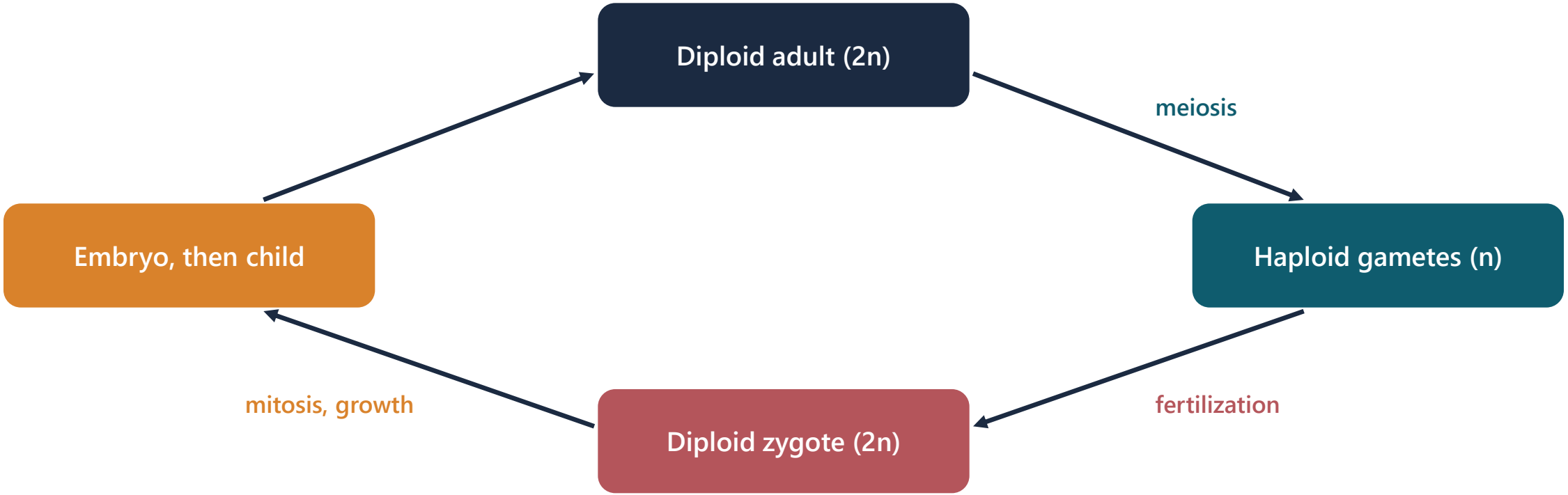


Example

Aphids reproduce asexually in summer, when conditions are stable, and sexually in autumn. Image: Lucis, public domain.

Meiosis and fusion in the life cycle

D3.1.2



Common mistake

Haploid means one set of chromosomes, not "23 chromosomes": 23 and 46 are human numbers only. Without meiosis, chromosome number would double every generation.

How new combinations arise

D3.1.2

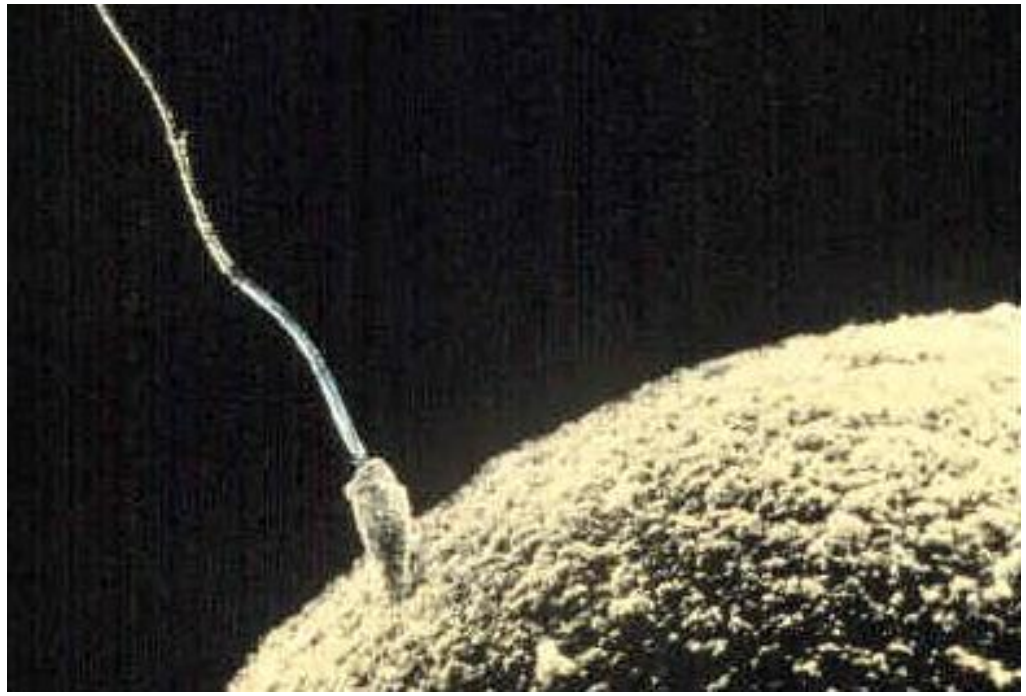
Stage	What it does to alleles
Meiosis	**Breaks up** the parents' combinations: each gamete is unique
Fusion of gametes	**Makes new** combinations: any sperm may fertilize the egg

Change and continuity

Genes pass from one generation to the next: continuity. Meiosis and random fusion shuffle them: variation for evolution.

Male and female gametes differ

D3.1.3



A sperm at the surface of an egg. Public domain.

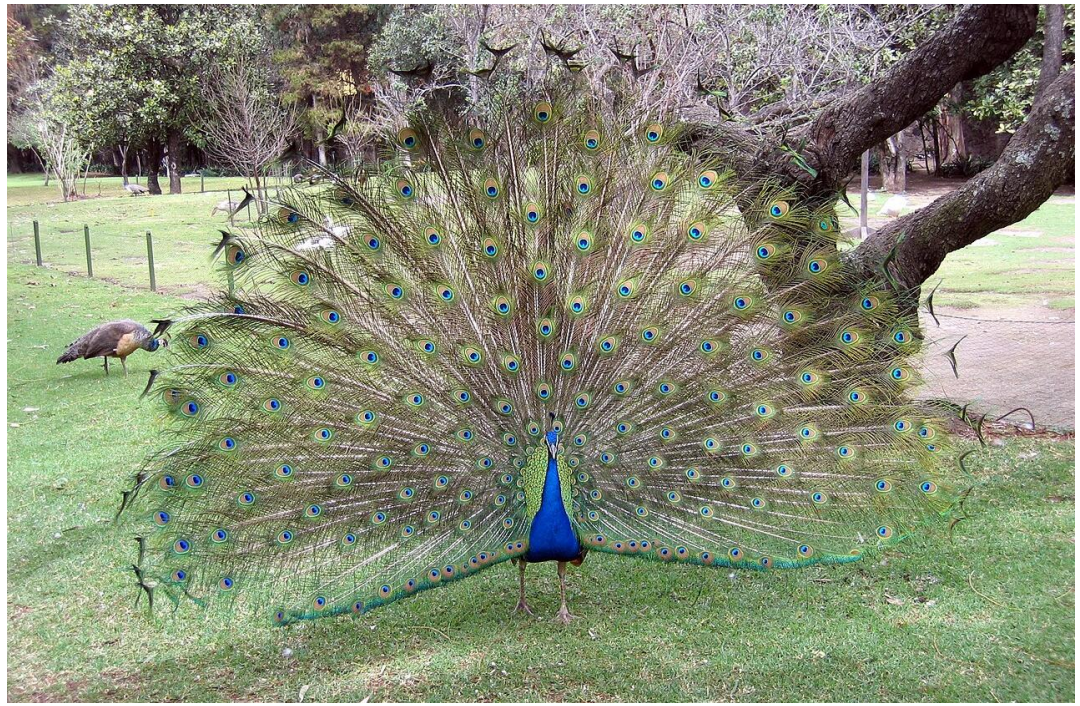
	Male gamete	Female gamete
Movement	Travels	Does not travel
Size	Very small	Much larger
Food reserves	Few	Many
Number	Very many	Few

Key idea

The male gamete **travels** to the female one. So it is small and light, many are made because few arrive, and the egg carries the food.

Different reproductive strategies

D3.1.3



Peacock display. Peter Andersen, CC BY 2.5.

In many animals, males make huge numbers of gametes and can mate often; females make fewer, larger gametes and invest more in each offspring.

So in many species males compete and females choose. These are patterns, and vary widely.

Discussion

What can biology tell us about animal behaviour, and what can it not tell us about human roles?

Practice: which kind of reproduction?

D3.1.1

Aphids in June

Plenty of food and a stable climate.

Coral reef warming

The sea is becoming warmer each decade.

Weed on new ground

A single seed lands on bare, empty soil.

For each one

Predict whether asexual or sexual reproduction is more useful, and explain using the words variation, adapted and environment.

Check-in: class 1

D3.1.1–3

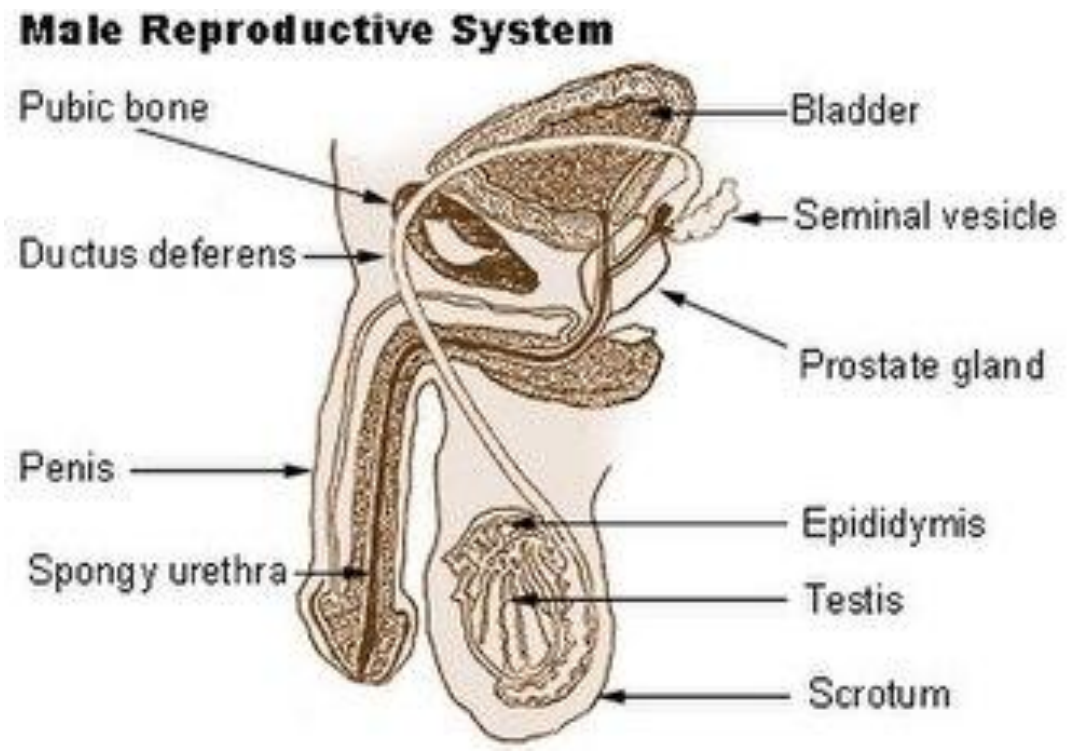
1. Give one advantage of asexual and one of sexual reproduction.
2. Explain why meiosis is needed in a sexual life cycle.
3. Explain how meiosis and fusion produce new combinations of alleles.
4. State the main difference between male and female gametes.
5. Explain why far more male gametes are produced.

How to use this

Answer without your notes, then check against slides 3–10.

The male-typical reproductive system

D3.1.4

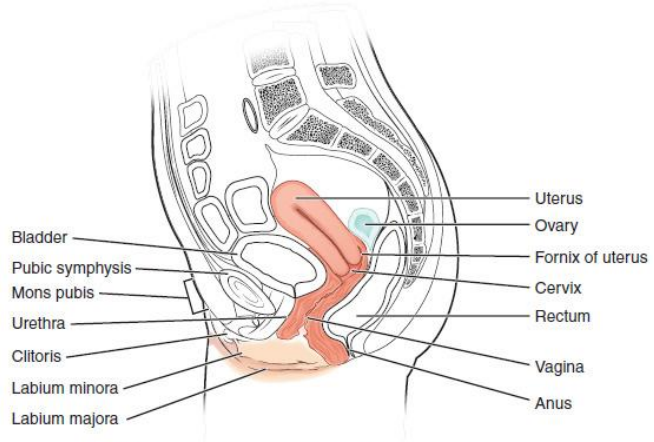


Male reproductive system. National Cancer Institute, public domain.

Structure	Function
Testes	Make sperm and testosterone
Scrotum	Keeps testes below body temperature
Epididymis	Stores sperm
Sperm duct	Carries sperm to the urethra
Seminal vesicles, prostate	Add fluid: nutrients, alkali
Urethra	Carries semen (and urine)
Penis	Erectile tissue for intercourse

The female-typical reproductive system

D3.1.4



Female reproductive system. OpenStax College, CC BY 3.0.

Structure	Function
Ovaries	Make eggs, oestradiol and progesterone
Oviducts	Carry the egg; site of fertilization
Uterus	Muscular wall; embryo develops here
Endometrium	Lining rich in blood vessels
Cervix	Connects uterus to vagina
Vagina	Muscular tube to the outside

Exam note

In a drawing, each oviduct opens into the space inside the uterus.

Four hormones control the cycle

D3.1.5

Hormone	Made by	Main role
FSH	Pituitary gland	Follicles develop and secrete oestradiol
LH	Pituitary gland	Triggers ovulation; forms the corpus luteum
Oestradiol	Developing follicle	Builds the endometrium; regulates FSH and LH
Progesterone	Corpus luteum	Maintains the endometrium; inhibits FSH and LH

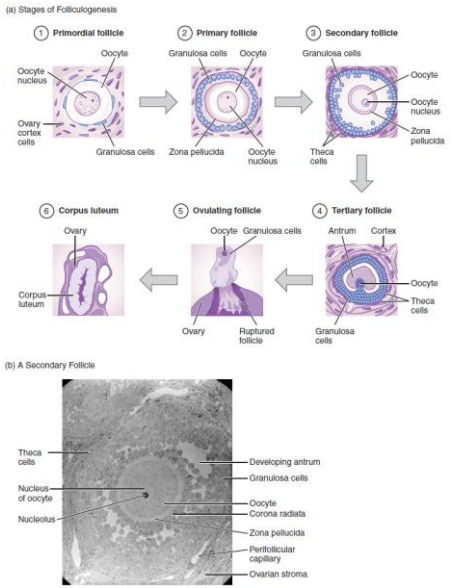
Two linked cycles

The **ovarian** cycle prepares and releases an egg. The **uterine** cycle builds the lining, then sheds it if no embryo implants. Each takes about 28 days.

Days 1–14: preparing an egg

D3.1.5

Day 1: **menstruation** → **FSH** stimulates follicles → Follicle secretes **oestradiol**: lining rebuilds → High oestradiol: **LH surge** → Ovulation, about day 14



Stages of follicle development. OpenStax College, CC BY 3.0.

Oestradiol raises FSH receptors on the follicle, so it makes even more oestradiol: **positive feedback**.

High oestradiol also inhibits FSH, so no more follicles develop: **negative feedback**.

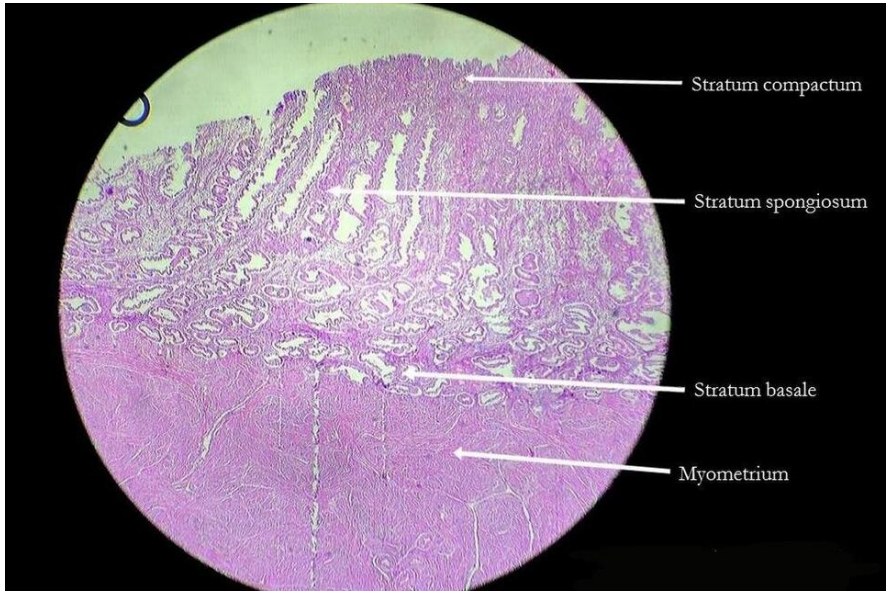
Common mistake

FSH: follicles grow. LH: ovulation. Do not swap them.

Days 14–28: ready for implantation

D3.1.5

LH forms the **corpus luteum** → It secretes **progesterone** → Lining kept thick → Progesterone inhibits FSH and LH → Corpus luteum breaks down; lining shed



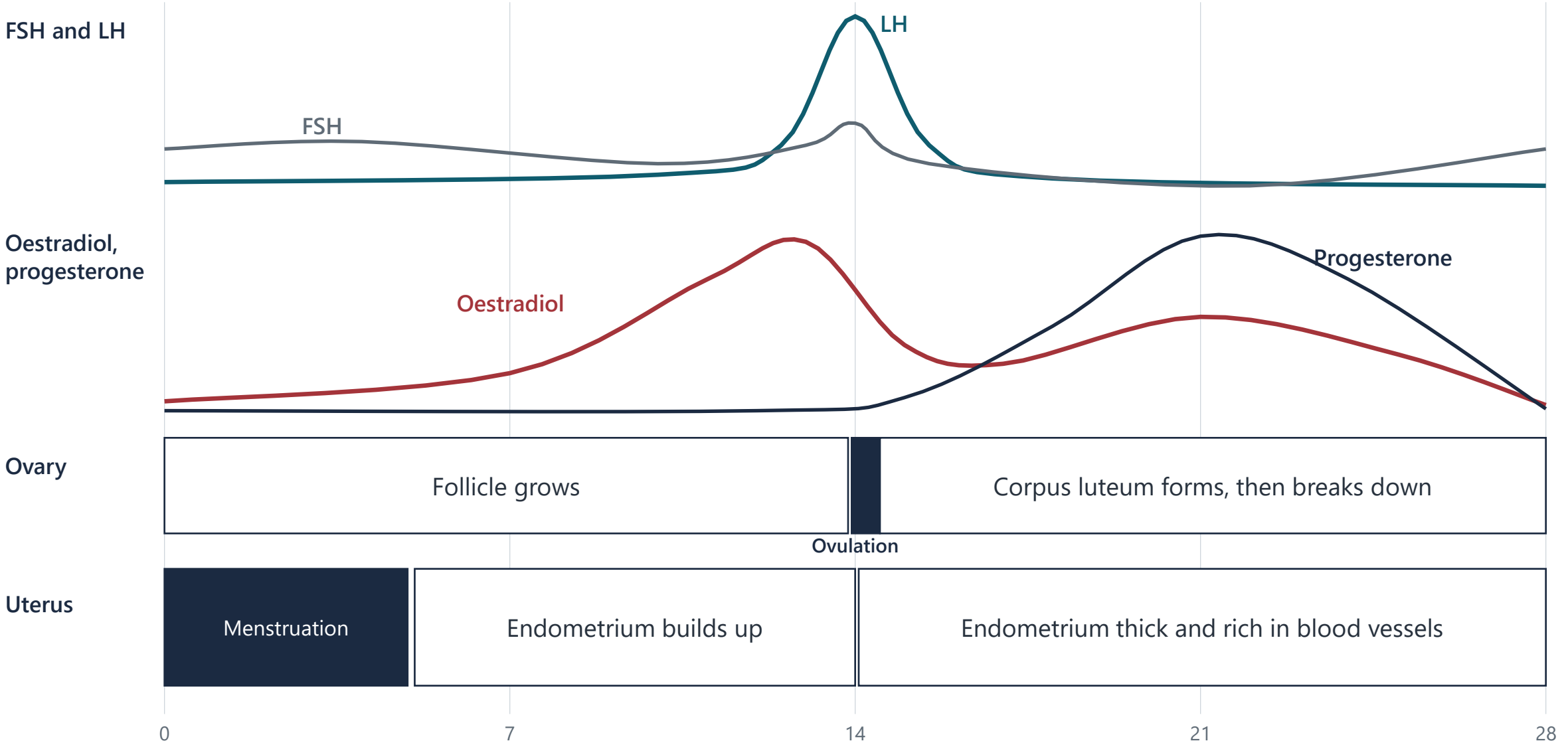
Endometrium, second half of the cycle. Dr. Roshan Nasimudeen, CC BY-SA 3.0.

With FSH and LH low, the corpus luteum is not maintained. Progesterone and oestradiol fall, the endometrium breaks down and a new cycle begins.

If an embryo implants, it secretes a hormone that keeps the corpus luteum going.

The cycle in one picture

D3.1.5



Positive and negative feedback

D3.1.5

Type	Example	Effect
Positive	Oestradiol raises FSH receptors on the follicle	Oestradiol rises further
Negative	High oestradiol inhibits FSH	No more follicles develop
Negative	Progesterone inhibits FSH and LH	No new ovulation; corpus luteum not maintained

Careful

When progesterone falls, FSH can rise again. That is the removal of negative feedback, not positive feedback.

Working with data: hormones in one cycle

D3.1.5

Relative blood levels of four hormones on selected days of a 28-day cycle. Values are invented for practice; the pattern is realistic.

Day	FSH	LH	Oestradiol	Progesterone
2	6	5	2	1
10	5	6	7	1
14	9	40	8	1
21	2	4	5	15
27	3	3	2	2

1. Identify the day of ovulation, with evidence.
2. Explain the high progesterone on day 21.
3. Explain why FSH is low on day 21.
4. Predict progesterone if an embryo implanted.

Check-in: class 2

D3.1.4–5

1. State the function of the epididymis and of the oviduct.
2. Name the hormone that triggers ovulation and the gland that makes it.
3. Describe the role of the corpus luteum.
4. Give one example of positive and one of negative feedback in the cycle.
5. Explain why the endometrium is shed at the end of a cycle.

How to use this

Answer without your notes, then check against slides 12–19.

Fertilization in humans

D3.1.6

Sperm meets egg in the **oviduct** → Membranes **fuse**; sperm nucleus enters → Sperm tail and mitochondria destroyed → Both nuclear membranes break down → Chromosomes join one mitosis: two diploid nuclei



Human embryo, eight-cell stage. RWJMS IVF Program, public domain.

Common mistake

The two nuclei do not simply merge. Their membranes break down and the chromosomes take part in a shared mitosis. All the mitochondria in the embryo come from the egg.

In vitro fertilization



Injecting a sperm into an egg. Eugene Ermolovich, CC BY-SA 3.0.

In vitro: in glass, outside the body. Used when natural fertilization fails, for example too few sperm or blocked oviducts.

It relies on **hormones**: the natural cycle is paused, then doses produce many eggs at a chosen time.

History

The first baby born by IVF arrived in 1978.

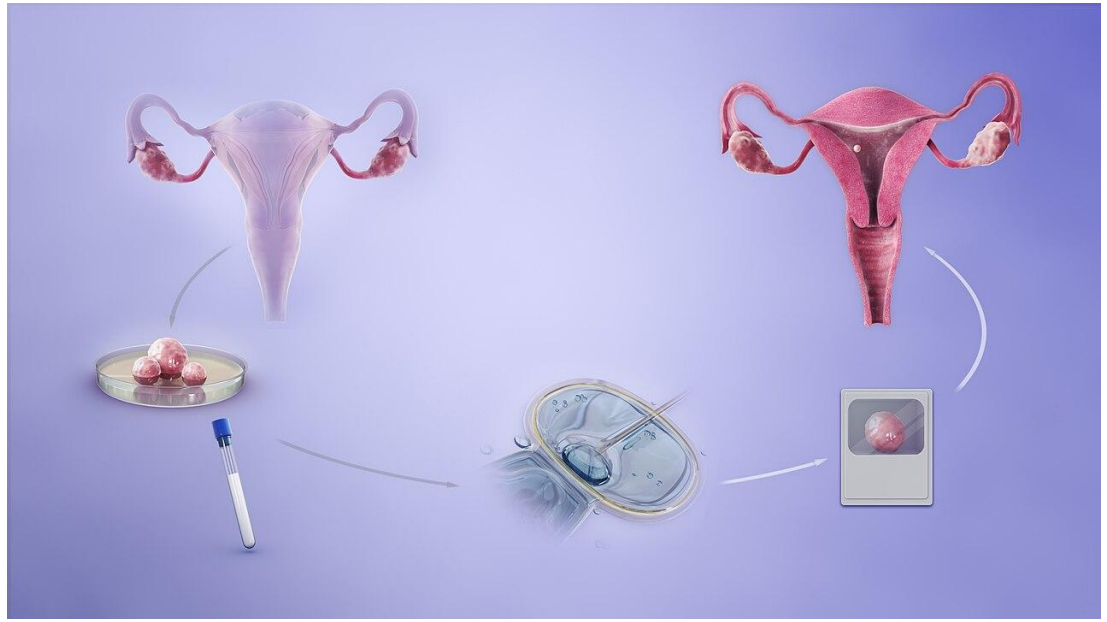
How hormones are used in IVF

D3.1.7

Stage	What is given	Why
Down-regulation	A drug for about two weeks	Stops the natural hormones, so timing is controlled
Superovulation	High doses of FSH, about ten days	Many follicles develop at once
Maturation	An hCG injection	Eggs mature; collected about 36 hours later
Fertilization	No hormone	Eggs mixed with sperm in a dish
Implantation	Progesterone	Keeps the endometrium thick

One or two embryos are then placed in the uterus.

Evaluating IVF



Stages of IVF. scientificanimations.com, CC BY-SA 4.0.

Not every treatment leads to a birth, and success falls with age. Treatment can be physically and emotionally demanding.

Several embryos raise the risk of twins or triplets; spare embryos raise ethical questions.

Discussion

Which of these questions can science answer, and which need other ways of thinking?

Practice: IVF success and age

D3.1.7

Live births per embryo transfer by age of the woman. The values are invented for practice; the pattern is realistic.

Age / years	Live births / %
Under 35	32
35–37	25
38–40	16
41–42	8
Over 42	3

1. Describe the trend.
2. How many times more likely is success under 35 than at 41–42?
3. Suggest a biological reason for the trend.
4. Suggest why many clinics now transfer only one embryo.

Check-in: class 3

D3.1.6–7

1. State where fertilization usually happens.
2. Outline what happens to the nuclei at fertilization.
3. Explain why down-regulation is used in IVF.
4. Explain why FSH is injected during IVF.
5. Give two arguments for and against IVF.

How to use this

Answer without your notes, then check against slides 21–25.

Common mistakes to avoid

Not this	But this
FSH causes ovulation.	LH causes ovulation; FSH makes follicles grow.
Oestradiol is made by the pituitary.	It is made by the developing follicle.
Menstruation happens because the egg dies.	It happens because progesterone falls.
In IVF, fertilization happens in the uterus.	It happens in a dish; embryos then go to the uterus.
Haploid means 23 chromosomes.	Haploid means one set of chromosomes.

Summary: reproduction

D3.1.1–7

	Key idea
D3.1.1	Asexual: identical offspring. Sexual: varied offspring
D3.1.2	Meiosis halves; fusion restores; both make new combinations
D3.1.3	Male gamete travels: small, many; female gamete: large, few
D3.1.4	Testes, sperm duct, urethra; ovaries, oviducts, uterus
D3.1.5	FSH, oestradiol, LH surge, progesterone; feedback
D3.1.6–7	Fusion in the oviduct; IVF uses hormones to control timing

Check that you can...

1. **Compare** sexual and asexual reproduction
2. **Explain** the roles of meiosis and fertilization
3. **Distinguish** between male and female gametes
4. **Annotate** diagrams of the reproductive systems
5. **Explain** the hormonal control of the menstrual cycle
6. **Outline** fertilization in humans
7. **Explain** and **evaluate** the use of hormones in IVF

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one advantage of asexual reproduction and one of sexual reproduction.

2. Which hormone triggers ovulation? Which structure secretes progesterone after ovulation?

3. Why does the endometrium break down at the end of a cycle with no pregnancy?

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